

An ADLINK Industry White Paper

Vortex OpenSplice Data Distribution Service (DDS) Professional Services

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First specified in 2003 by the OMG, the Data Distribution Service (DDS) introduced the concept of data-centric real-time publish/subscribe, a powerful means for the ubiquitous and real-time sharing of information by distributed applications. Exploiting this technology requires the use of sophisticated middleware and tools such as Vortex™ OpenSplice™ from ADLINK, as well as a good grasp of the fundamentals of the specification and its application for complex, distributed networked systems. Implementing open and interoperable architecture solutions for such demanding system environments is therefore challenging, and subject to multiple bottlenecks if not carefully designed and deployed throughout the whole system lifecycle.

ADLINK has a wealth of experience in the design and development of data-centric mission-critical systems, using both CORBA and DDS technologies. ADLINK provides the expertise in architecting and developing such systems as well as the middleware products and modelling tools that allow you to implement them. Conducted by our DDS experts, our broad suite of Professional Services Packages will speed up the entire development process and ultimately provide you with a faster time to market.

ADLINK Vision and Value Add

World-class expertise and best-practice project management (Solution9001SM methodology) is encapsulated in a broad range of professional services from our DDS experts: training, workshops, development, and consulting.

The recognized leader of the QoS-based Service Oriented Architecture (SOA) technology field, ADLINK's know-how of DDS and its field implementation is second to none:

- **Access to Unique Technology Expertise** through World Class DDS Experts and key contributors to the OMG specification process for DDS and the definition of standards-based solutions.
- **Access to Unique Field Expertise**, through our consultants involved for over a decade in the successful design and implementation of complex mission-critical systems in combat management, air-traffic control management and transport system monitoring fields. Such knowledge is manifold, including:
 - » Information modeling: proper design of the data sharing model (domains, topics, etc.) best adapted to the specifics of our Customer's system requirements.
 - » Data transport optimization through the definition of the most appropriate QoS to implement.
 - » Proper integration with other distributed technologies such as CORBA, OO Databases and RTOS systems.
- **Solutions based on the most complete DDS product:** Vortex™ OpenSplice™ is the most complete implementation of the DCPS and DLRL specifications, thus guaranteeing high-performance, scalable, reliable and secure data distribution for LAN as well as WAN based networked systems.
- **Proven track record for on-time, on-budget and on-quality completion** of our professional services through our Solution9001SM methodology.

Expertise

ADLINK's DDS expert team has a wealth of experience building or supporting the development of mission-critical systems in multiple domains. Examples include:

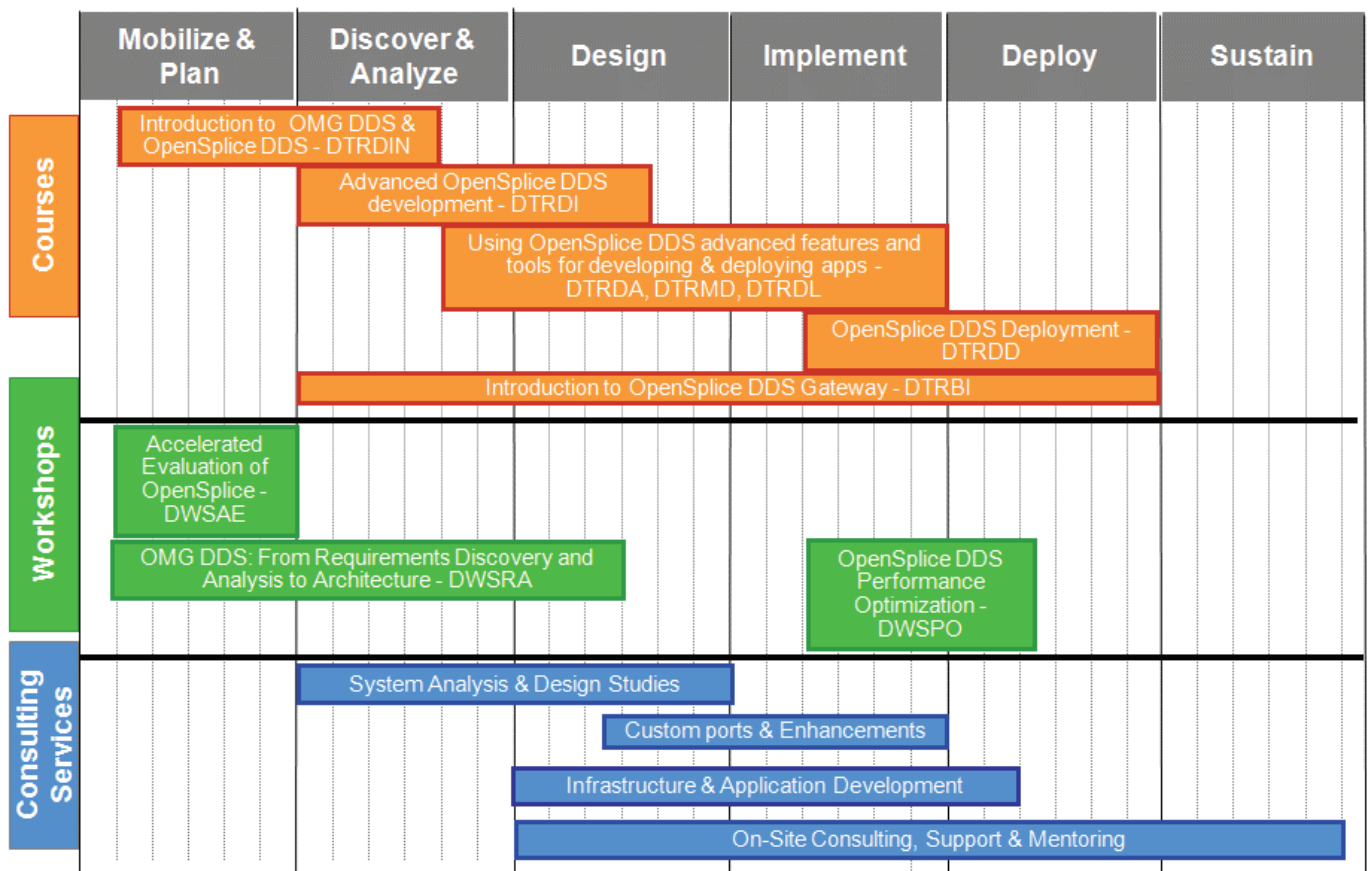
- Multiple Naval combat systems
- Various Army systems
- Tactical networks
- Public transport infrastructure systems
- Air Traffic Management systems
- Space programs
- Military communications systems
- Satellite communications systems
- Rocket launch installation communication systems
- Complex Medical devices
- Agricultural Vehicle Systems
- Commercial Vetronics
- Smart Grid Architectures
- Train Control Systems
- High Frequency Auto-Trading

Capabilities

Conducted by our DDS experts, we offer a broad spectrum of DDS training, workshops and consulting services supporting all phases of a project development and deployment:

- Training is meant to properly introduce the DDS specification as well as the use of Vortex™ OpenSplice™ middleware and tools.
- Workshops are meant to resolve DDS field design and implementation issues through short focused consulting sessions and the delivery of a recommendation report.
- Consulting services include a broad spectrum of additional services from mentoring and on-site support to the development of parts of the Customer DDS-based infrastructure and system integration.

Our offerings are visualized in the picture below and summarized one by one in this section:



Benefits

Based on its broad DDS vision and experience, ADLINK brings key benefits that will maximize the chances for a successful and effective implementation of DDS in mission-critical systems. The main advantages of using ADLINK's combined product and services suite include:

- **Improved Productivity:** with effective DDS technology transfers and provision of proper expertise at key project phases from design to implementation.
- **Risk Reduction:** defining the right system architecture, thus ensuring a scalable and reliable system behavior, with the help of our DDS experts and solution-oriented workshops.
- **Shorter Time to Market:** speeding up the entire development process and ultimately ensuring a faster time to market and significant cost reduction.
- **Improved Quality:** Learn from our experts and apply data-centric and QoS-enabled SOA best practices with the fully customized workshop approach.
- **Cost Reduction:** through the global optimization of the usage of DDS and Vortex™ OpenSplice™ as the information backbone of your networked system applications.

For More Information

For further information regarding Vortex OpenSplice, please e-mail: ist_info@adlinktech.com or visit: www.adlinktech.com

